



RETRO-FIT
COCKPIT DOOR
SURVEILLANCE
(CDSS)

INTRODUCTION



REGULATIONS



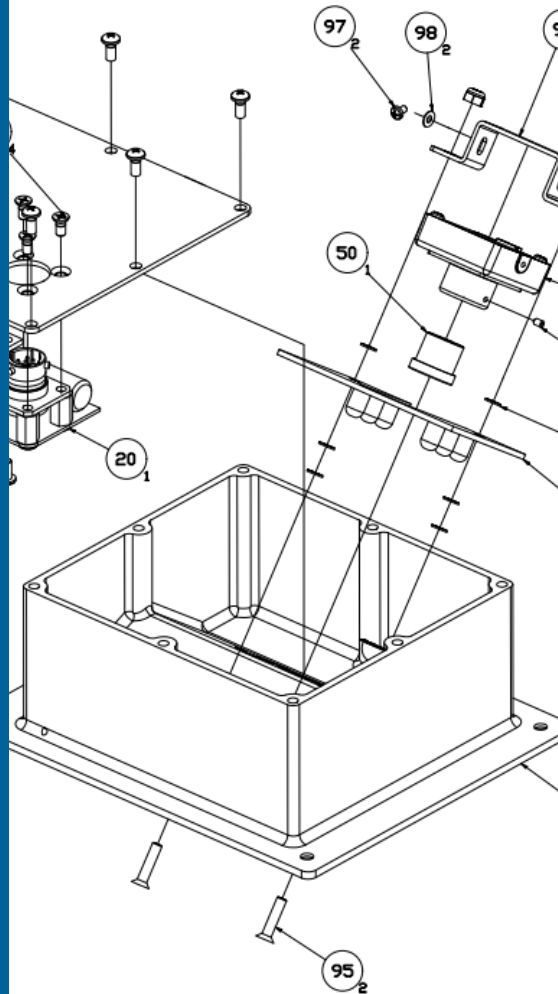
ICAO



EASA
European Union Aviation Safety Agency



TECHNICAL



MATERIALS





INTRODUCTION

Satisfies the requirement for a non-intrusive, non-distracting method of identification by the Captain and First Officer of who is trying to gain access to the Flight Deck.

Acts as deterrent to terrorism and air rage and helps to prevent catastrophic loss of aircraft from terrorism or revenge motivation.

Represents low-cost option to significantly increase levels of aircraft safety and security.



OFFERABILITY

BOEING 737 CLASSIC
BOEING 737 NG
AIRBUS A320 CEO
AIRBUS A320 NEO
DHC DASH 8 Q400
ATR 72
FOKKER 70
FOKKER 100

INSTALLATION

EASA STC
EASA MINOR MODIFICATION
CAA MINOR MODIFICATION



REGULATIONS

Global regulators strongly recommend video systems...

Many European carriers proceeded with the installation of cockpit door video monitors shortly after 9/11 and before mandates were issued.

ICAO Annex 6 – Part 1 – Chapter 13 – Standard 13.2.3(b) (1 Nov 2003):

“means shall be provided for monitoring from either pilot’s station the entire door area outside the flight crew compartment to identify persons requesting entry and to detect suspicious behaviour or potential threat.”

...interpreted into the Eurocae Minimum Operating Specification ED-123 which describes the minimum specification for a cockpit door surveillance system.

EU COMMISSION REGULATION (EC) No 859/2008 Sub-Part S OPS 1.1255 (20 Aug 2008):

“In all aeroplanes which are equipped with a flight crew compartment door...means shall be provided for monitoring from either pilot’s station the area outside the flight crew compartment to the extent necessary to identify persons requesting entry to the flight crew compartment and to detect suspicious behaviour or potential threat.”



TECHNICAL

System typically features:

3x *AD Aerospace* FV-0410 Internal Monochrome Video Cameras

1x *AD Aerospace* FV-0835 LCD Monitor mounted in the Flight Deck Centre Console (an additional monitor is provided for the Airbus A320)

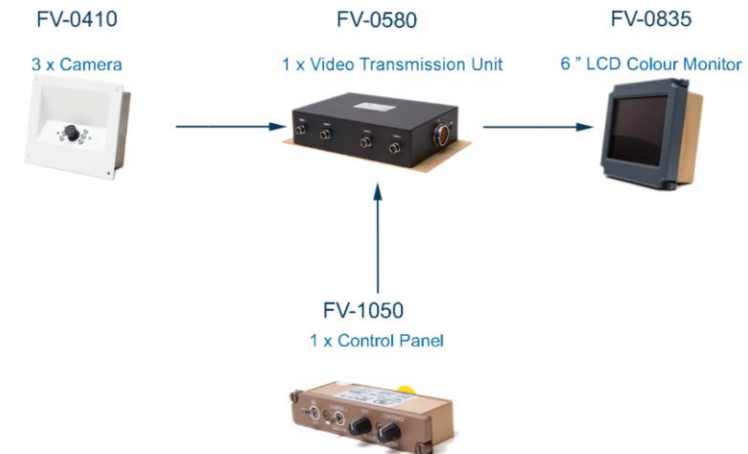
1x *AD Aerospace* FV-1050 Control Panel also mounted in the Flight Deck Centre Console (an additional Control Panel is required for the Airbus A320)

1x *AD Aerospace* FV-0580 Camera Interface Unit.

EASA approved complete installation kit manufactured by *AD Aerospace*.

Cameras are positioned strategically outside of the Flight Deck door to give full coverage of the galley area without blockage or interference.

Retro-Fit Cockpit Door Surveillance Solution (CDSS/FDEVSS)



MATERIALS

FV-0410 CAMERA



FV-0580 VIDEO SWITCH



FV-0835 MONITOR



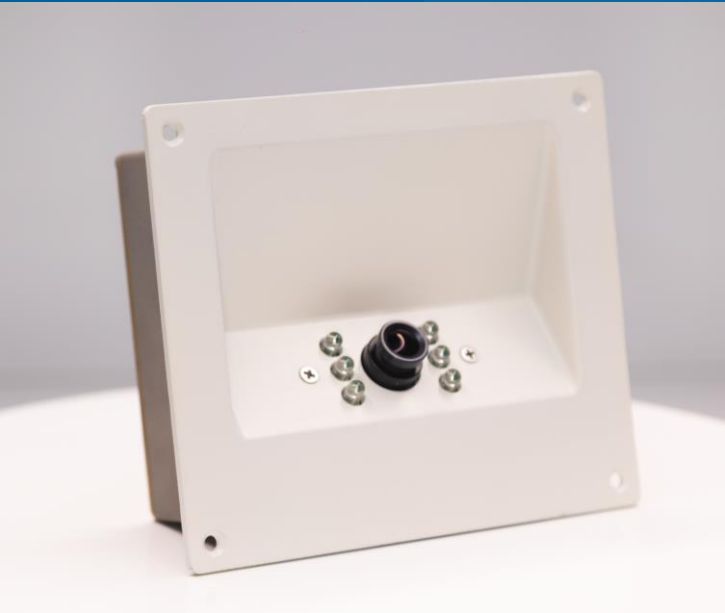
FV-1050 CONTROL PANEL



MATERIALS



FV-0410 CAMERA



AD Aerospace’s FV-0410 Internal Aerospace Camera is designed for use within the cabin of a commercial aircraft, to provide the overt but non-obtrusive monitoring of areas including the cockpit approach.

The camera is monochrome to provide the clearest image and best contrast in the hugely variable light conditions found in aircraft flying at 50,000 feet.

The Infra-Red (IR) Illuminators produce light at an 875nm wavelength which is invisible to the human eye but within the sensitivity of the camera. This allows clear images to be produced in low light situations, such as night-time light levels on an aircraft, and helps compensate when a subject is strongly backlit.

The units have a variety of lens options and colour finishes to suit most requirements.

Specifications	
Size	140mm x 120mm x 52mm
Weight	425g Maximum
Video Output	1.0V p-p into 75R composite video (unbalanced), 50R or 100R balanced video as an option.
Sensitivity	Better than 0.01 Lux at f2.0 6 LEDs, at +/- 30 degrees cone, at 875nm
Power	80mA from 28V source
Light Control	Auto electronic shutter to 1/100,000 seconds
Field of View	Various lenses from 26 to 135 degrees
Material	Cast Parts: Aluminium Alloy to LM25 Fabricated Parts: Al Alloy to BSEN485:3103H4/5251H22, or 6061T4
Paint Finish	Alocrom to Def Stan 03-18 Finish to suit cabin interior.
Resolution	Greater than 420 TV Lines per picture height (horizontal) in picture centre. Greater than 300 lines in the corners.
MTBF	50,000 hours

MATERIALS



FV-0835 MONITOR



AD Aerospace's FV-0835 is an airborne 6" LCD monitor designed to display images from external and/or internal cameras installed on board an aircraft.

It is light weight, low powered, and can be quickly installed/maintained as it is Dzus rail mounted.

The design is intended primarily to satisfy the requirement for a monitor to display images from cameras mounted on a Boeing 737 aircraft (located centre console). The unit is, however, to be designed in such a way that it may be applicable to a wider range of aircraft.

Specifications	
Dimensions	Shallow Back: 146mm x 133mm x 67mm
Weight	900g Maximum
Input	Single composite video: PAL or NTSC standard
Screen Size	5.6-inch Diagonal
Display Resolution	960 x 234 pixels
Power	The unit runs from 28V power The power drawn is a maximum of 5W
Connectors	Mil-C-38999 connector
Environmental Qualification	RTCA DO160E\ED-14E, for cockpit mounted equipment
MTBF	25,000 hours (excluding LCD backlight)

MATERIALS



FV-0580 VIDEO SWITCH

AD Aerospace’s FV-0580 Aerospace Video Switch Unit takes input from up to 4 analogue cameras and produces video streams suitable for viewing on displays, such as LCD monitors.

The unit is light and small enough to be fitted in cabin, overhead.



Specifications	
Dimensions	275mm x 160mm x 60mm
Weight	1kg Maximum
Input	Video from up to 4 cameras each on its own connector, PAL or NTSC standard. The unit caters for both unbalanced and balanced video.
Screen Size	5.6-inch Diagonal
OSD function	Enables text to be defined and superimposed on the video usually to indicate camera view.
Power	Maximum under normal operating conditions of 0.5A at 28V to aircraft RTCA Do160D.
Connectors	Mil-C-38999 connectors
Environmental Qualification	RTCA DO160E\ED-14E, for cockpit mounted equipment
MTBF	50,000 hours

MATERIALS



FV-1050 CONTROL PANEL



AD Aerospace’s FV-1050 is designed to be fitted in the cockpit instrument panel of a commercial aircraft.

The front panel switches are fitted behind a "Light plate" which gives trans-illumination through the knobs, around the panel, and through the lettering, allowing the switches to be identified in a darkened cockpit.

Power for the “Light plate” is provided from a 5V ac source external to the unit.

There is no software component in the unit.

Specifications	
Dimensions	146mm x 28.5mm x 71.2mm
Weight	0.25kg Maximum
Material	Box: 1.2mm Al Alloy sheet BSEN 485:4303H14/5251H22 Backplate: BSEN 485: 6082T6/6262T6-T9 Light plate: Acrylic
Connectors	Mil-C-38999 connector
Environmental Qualification	RTCA DO160E\ED-14E, for cockpit mounted equipment
MTBF	100,000 hours

RETRO-FIT CDSS CUSTOMERS



Dubai Aerospace Enterprise



الخطوط الملكية المغربية
royal air maroc



INSTALLATION EXAMPLES



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